# SMOLSolver: Lightweight Generator – Verifier Framework for Math Problem Solving

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## Background and Motivation

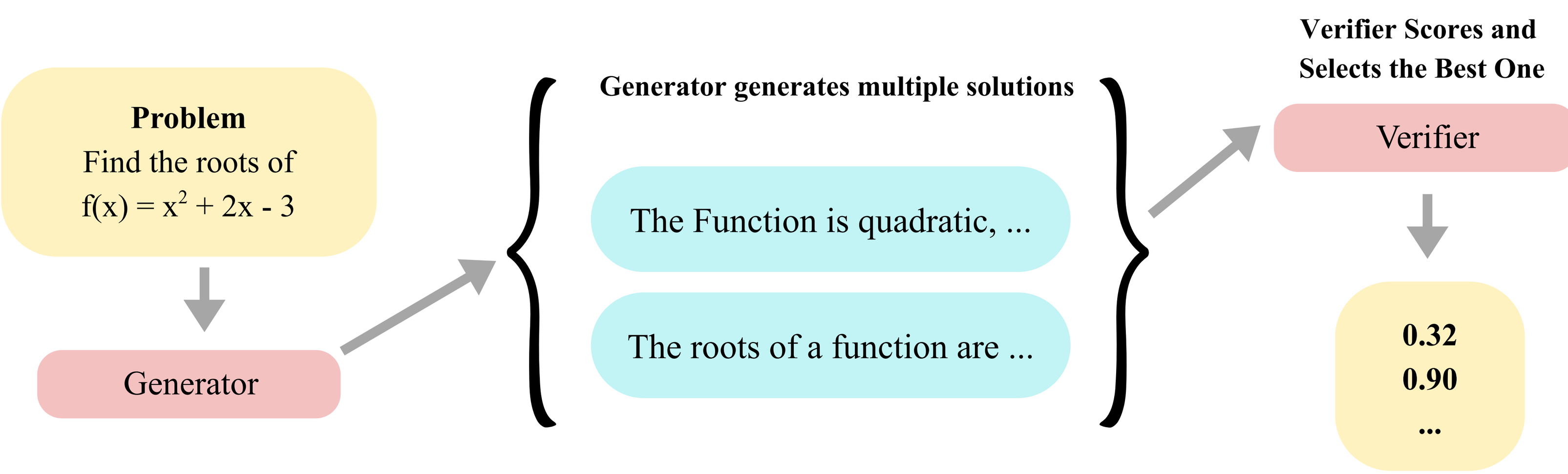


Figure1: Verfier enhanced self consitency for improving generator outcome

## Overview

We introduce SMOLSolver, a modular Generator–Verifier–Ranker (G–V–R) pipeline for arithmetic problem solving on GSM8K.
Given an input problem, the framework produces a set of sampled reasoning solutions, evaluates their correctness, and selects a final answer using various aggregation rules.

The system consists of:

- **Generator G:** a LoRA-fine-tuned Phi-2 model producing sampled solutions.
- **Verifier V:** a RoBERTa-large binary classifier predicting correctness of a candidate solution.
- **Ranker R:** a RoBERTa-large regression model assigning continuous quality scores.

## Method

We introduce G-V-R pipeline in SMOLSolver. Used 1319 test questions in test phase, our approach consist of two key parts:

**Aggregation and Evaluation Methods:** For each question, the Generator produces 15 sampled solutions. Each sample(s) consist of reasoning text (c), and extracted numeric answer(a). We applied 7 methods utilizing the combination of G, V, and R.

| Method | Equation |
|---|---|
| Majority Vote (MV) | $a_{\text{MV}} = \arg\max_a \text{freq}(a)$ |
| G + Classifier (Top) | $a_{\text{V-top}} = a_j, \quad j = \arg\max_i V(s_i)$ |
| G + Classifier (Weighted) | $a_{\text{V-w}} = \arg\max_a \sum_{i:a_i=a} V(s_i)$ |
| G + Ranker (Top) | $a_{\text{R-top}} = a_j, \quad j = \arg\max_i R(s_i)$ |
| G + Ranker (Weighted) | $a_{\text{R-w}} = \arg\max_a \sum_{i:a_i=a} R(s_i)$ |
| G → Classifier → Ranker (Top) | $a_{\text{C→R-top}} = a_{\arg\max_i R(s_i)}$ |
| G → Classifier → Ranker (Weighted) | $a_{\text{C→R-w}} = \arg\max_a \sum_{i:a_i=a} (\alpha V(s_i) + (1-\alpha) R(s_i))$ |

Table1: Decision Rules for Generator–Verifier/Ranker Methods

**Generator Sampling and Representations:** For robustness, we estimate generator stability by computing answer-frequency counts:

$$r_1 = \max_a \ \text{freq}(a), \quad r_2 = \text{2nd largest freq}(a), \quad \text{margin} = r_1 - r_2$$

Two learned thresholds $\tau_{\text{low}}, \tau_{\text{high}}$ partition the test set into:

- **High-consensus**: margin $\geq \tau_{\text{high}}$
- **Medium-consensus**: $\tau_{\text{low}} <$ margin $< \tau_{\text{high}}$
- **Low-consensus**: margin $\leq \tau_{\text{low}}$

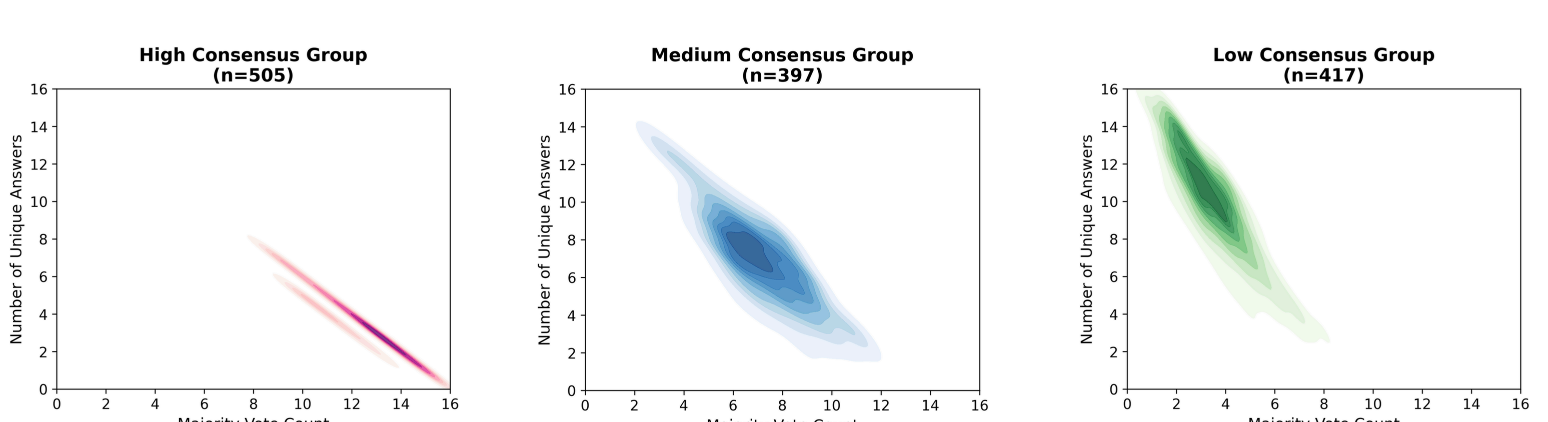


Figure2: Consensus-Level Clusters of Generator Outputs

High: margin > 0.533 → 505
Medium: 0.133 < margin ≤ 0.533 → 397
Low: margin ≤ 0.133 → 417

- To obtain balanced groups, we split the dataset at the 33rd and 66th percentile margins.
- This yields three similarly sized groups, enabling reliable cross-group comparison.

## SMOLSolver

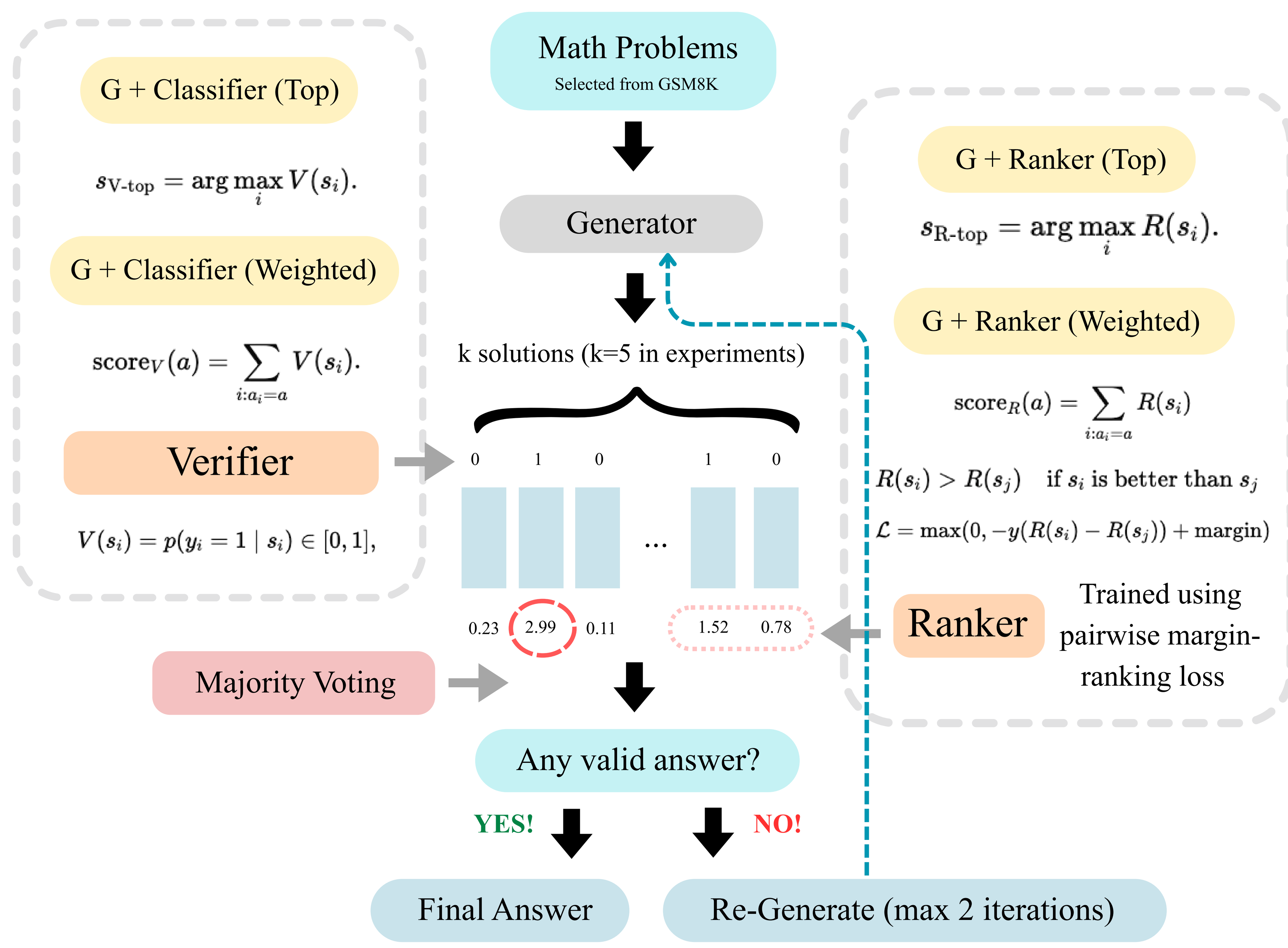


Figure3: Modular G–V–R Framework

## Experiments and Results

**Finding 1:** Failure of Verifier training on dataset with synthetic corruptions

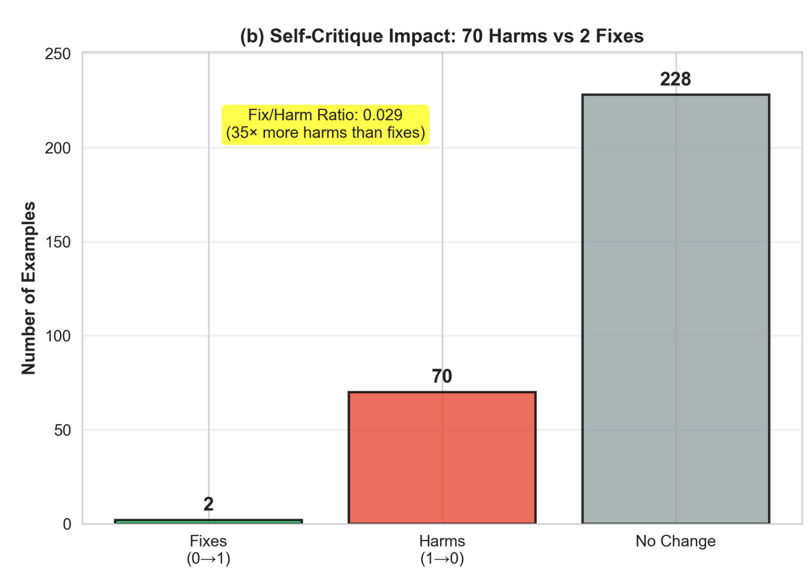


Figure4: Self-Critique Outcome Counts

**Finding 2:** Increasing of solutions make Generator perform better.

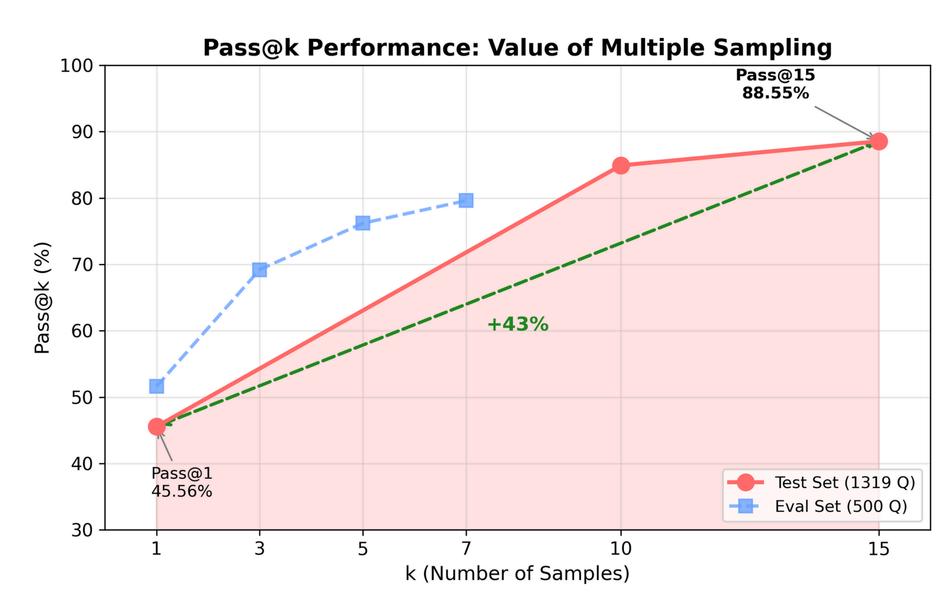


Figure5: Pass@k Gains vs. Number of Samples

**Finding 3:** Across all consensus levels, Majority Vote leads, but Ranker-Weighted uniquely improves the unstable medium-consensus subset without harming performance.

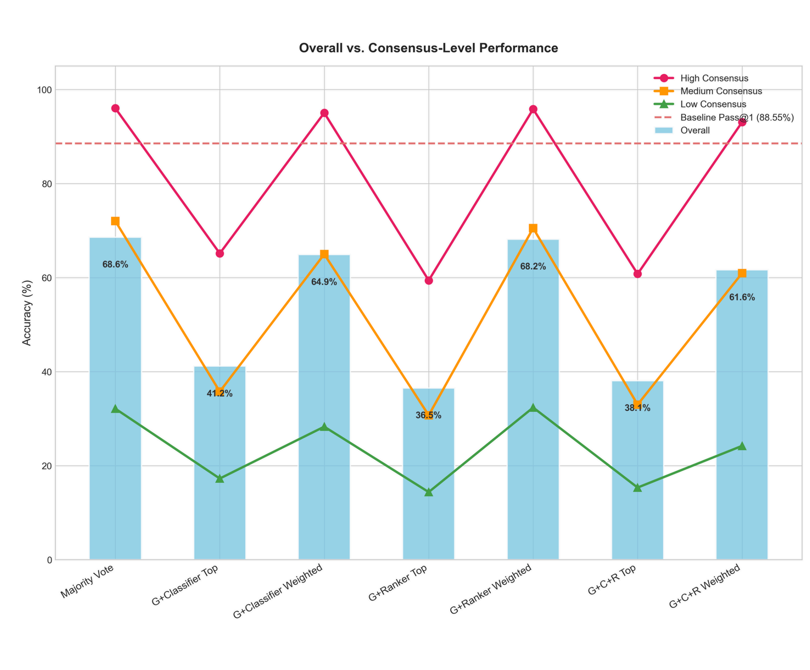


Figure6: Consensus-Level Accuracy Comparison

**Finding 4:** While Majority Vote generally performs well across consensus levels, the Weighted Ranker achieves the strongest performance specifically on the low-consensus subset.

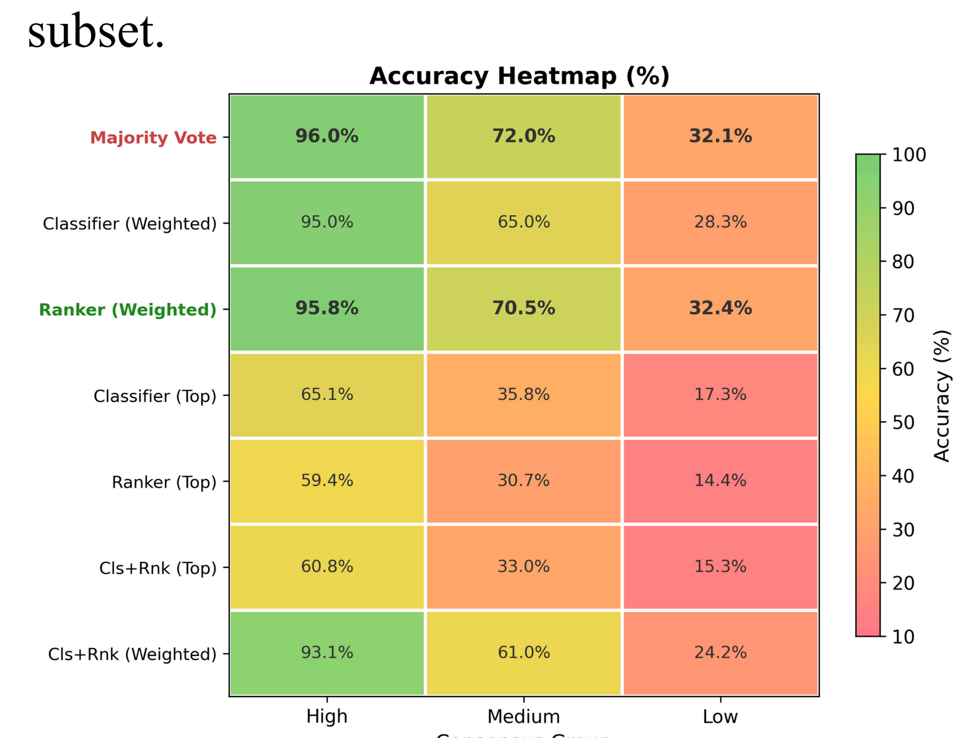


Figure7: Accuracy Heatmap Across Consensus Groups

**Finding 5:** Adaptive method achieves the best accuracy-to-compute tradeoff, fixing far more errors than it creates while preserving more than 70% neutral expansions. This is a smaller & independent trial, done on a smaller test subset, to demonstrate the idea of adaptive majority voting (increase value of k if low consensus).

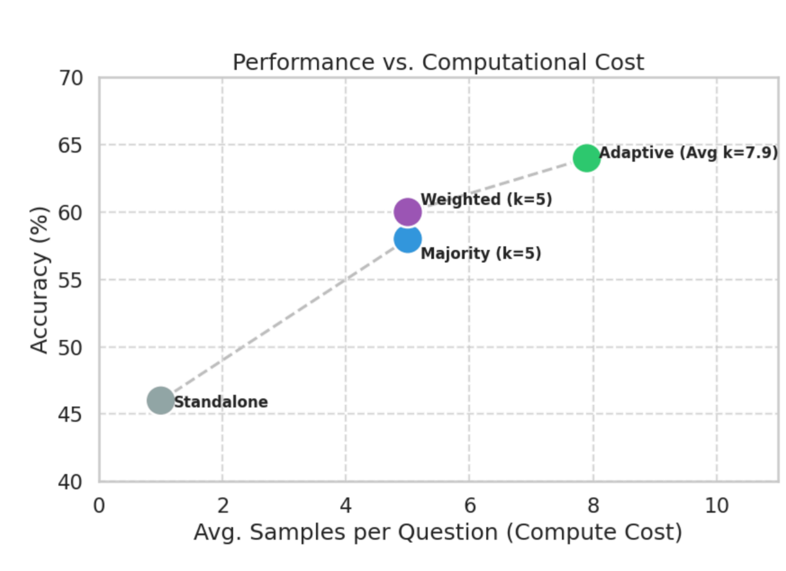


Figure8: Performance–Cost Trade-off

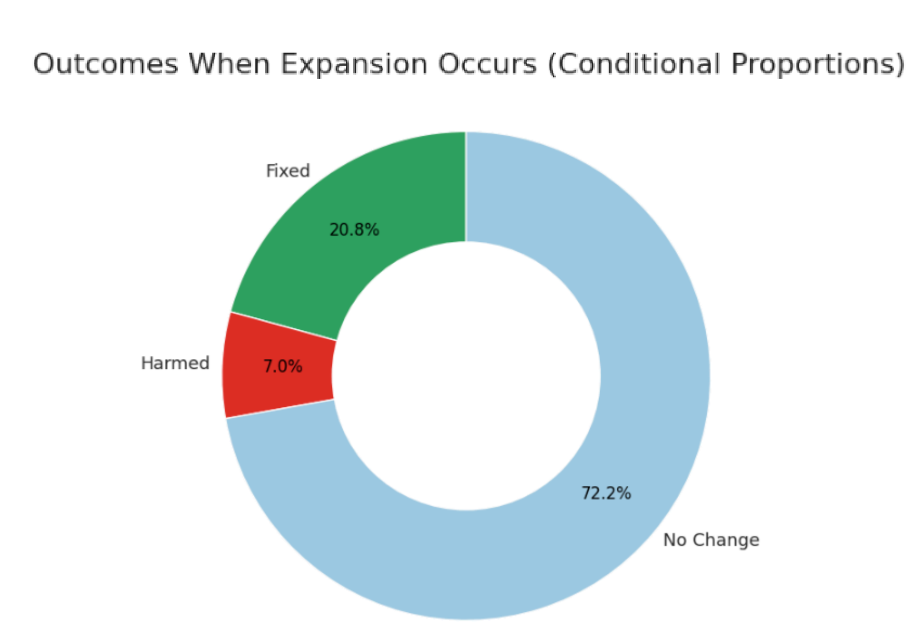


Figure9: Conditional Outcomes of Expansion